

Ethan Pratte

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Mechanical Engineering undergraduate at BYU combining industry experience in mechanical and electrical design with technical skills in SolidWorks, Python, C++, and manufacturing. I turn ideas into things that actually work.

EDUCATION

Brigham Young University - Ira A. Fulton College of Engineering

Dec 2027, Provo Utah

Mechanical Engineering Professional Program

- GPA: 3.6 / 4.0
- Active Participant in ASME (American Society of Mechanical Engineers)
- Relevant Coursework: CAD, Mechatronics, Dynamic System Modeling, Statics, Dynamics, Manufacturing, Mechanics of Materials, Electrical Engineering, Multivariable Calculus, Differential Equations, Linear Algebra

SKILLS

- *Software:* SolidWorks, PTC Onshape, Creo Parametric, Excel, Microsoft Office, Google Workspace
- *Programming:* Python (NumPy, Pandas, Scipy, Matplotlib), C++/C, PLC, ESP32, Arduino
- *Hardware/Lab:* Machine Design, DFM, GD&T, Machining, 3D Printing, Electrical Wiring/Design, Circuitry, PLC Wiring, Pneumatics, Lean Six Sigma, Welding, Soldering, Woodworking

PERSONAL PROJECTS

Quadruped Robot

- Designed and built a self-balancing 3D printed quadruped using 12 DoF and three 10.2kg-cm servo motors per leg. ESP-NOW for wireless remote control. Uses inverse kinematics with a custom gait algorithm for smooth walking.

Character Recognition Neural Network

- Built a neural network to recognize handwritten characters. Multiple layers of nodes with backpropagation calculus for weight updates. Trained on 60,000+ pregenerated images. Custom UI and normalization algorithm for preprocessing.

Quadcopter Drone Airframe

- Designed a foldable quadcopter in SolidWorks, modeling the airframe and mechanical subsystems. Folding mechanism enables compact storage. Built for carbon fiber construction as a larger camera drone.

3D Graphics Engine

- Built a 3D graphics engine using linear algebra for transforms and projection. Objects are 3D vertices, with matrices applied to translate, rotate, and scale in world coordinates. Camera moves/rotates via FOV for a navigable 3D scene.

WORK EXPERIENCE

US Synthetic (Oil & Gas)

Apr 2026 - Present, Orem Utah

Student Engineer

- Work on various engineering projects spanning mechanical and electrical design, automation, and controls integration. Design and assemble custom machinery for the production and processing of polycrystalline diamonds (PCD) used primarily in oil and gas extraction. Apply Lean Six Sigma principles to improve process efficiency. Utilize 3D printing for rapid prototyping and custom part fabrication. Write technical documentation for systems and procedures.
- *Automatic Media Blaster:* Redesigned and built an automatic media blaster from scratch. Increased blasting throughput by 300%. Wrote PLC electrical schematic and designed pneumatic system. Manufactured and sourced custom parts.
- *Pyrometer Calibrator:* Designed and built a pyrometer calibrator cart with mechanical components including shelving and doors, constructed from T-slotted rails. Assembled electrical system with PLC and HMI integration.
- *Brazing Inductor:* Assembled mechanical components of industrial sized inductor machine and built electrical PLC control box from schematics. Improved safety features to reduce operational risk.

Tourific, Inc. (Travel App)

Aug 2024 - Apr 2025, Provo Utah

Data Mapping Specialist

- Analyzed U.S. roadway and location data to improve route accuracy and ensure reliable navigation results
- Integrated and validated roadway/POI datasets into backend software to strengthen app performance and routing
- Maintained organized spreadsheets and databases for routes and locations to ensure consistent and accurate data

Vivint Smart Home (Home Security)

May 2025 - Aug 2025, Odessa Texas

Sales Representative

- Used prospecting, objection handling, and closing techniques to secure new accounts
- Interacted with a diverse range of customers, tailoring presentations to their needs
- Personally generated over \$90,000 in company revenue over 4 months